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Complicated foot ulcers and high treatment costs in rural type 2 diabetic patients in vietnam: impact of infections on hospital stays

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Abstract

Background This study aims to describe the clinical characteristics and assess treatment expenditures among patients with type 2 diabetes mellitus (T2DM) who developed foot ulcers in Vietnam.

Methods A cross-sectional, hospital-based study was conducted among 283 inpatients. Data collected included demographic information, medical history, glucose and HbA1c levels, ulcer severity, bacterial culture results, and direct medical costs.

Results The mean fasting plasma glucose, random glucose, and HbA1c were 8.87 mmol/L, 14.41 mmol/L, and 8.94%, respectively. Patients from rural areas had marginally higher treatment costs than those from urban areas (VND 10,154,000 vs. 9,444,300), though the difference was not statistically significant. The average direct medical cost per patient was approximately VND 10.5 million, with antibiotic therapy and hospitalization being the major contributors. Higher ulcer severity (Wagner grade ≥ 3) and infections—particularly with *Acinetobacter baumannii* and *Pseudomonas aeruginosa*—were significantly associated with increased costs. Multivariate analysis confirmed that ulcer grade ($p = 0.038$) and bacterial infection ($p = 0.002$) were independent predictors of higher cost.

Conclusions Poor glycemic control and infection are key factors increasing the economic burden of diabetic foot ulcers. Interventions targeting infection prevention and early management of ulcer severity may reduce treatment costs, particularly in resource-constrained settings.

Keywords Type 2 diabetes, Diabetic foot ulcer, Diabetes complications, Foot infections, Vietnam

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Background

Vietnam is one of the 38 countries and territories of the International Diabetes Federation Western Pacific (IDF WP) region. The WP region accounts for more than one-third (38%) of the total number of adults living with diabetes and has the third-highest prevalence of diabetes (11.9%) globally. The IDF projects that the number of people with diabetes in the WP region will increase by 27%, reaching 260 million by 2045, with a projected prevalence of 14.4% [1]. Type 2 diabetes mellitus (T2DM) is an escalating worldwide epidemic that contributes significantly to premature morbidity and mortality.

According to the International Diabetes Federation, the estimated incidence of diabetes was 9.3% (463 million) in 2019 and is projected to increase to 10.2% (578 million) in 2030 and 10.9% (700 million) in 2045 [2]. The prevalence of diabetes was greater in high-income countries than in middle- and low-income countries. Additionally, 50.1% of diabetic patients are undiagnosed [2]. In Vietnam, driven by economic growth and changing lifestyles, diabetes has emerged as a pressing public health concern. According to the IDF Diabetes Atlas 2021, the estimated diabetes prevalence is 6.1% (3.99 million people).

As a consequence, the burden of complications—particularly diabetic foot disease—is increasing. Diabetes elevates the risk of macrovascular complications such as coronary heart disease, stroke, and peripheral vascular disease, the latter being a primary cause of lower limb amputation. Simultaneously, microvascular complications such as retinopathy, nephropathy, and neuropathy exacerbate the risk of foot ulcers and infections. For severe foot complications, amputation is often necessary. The amputation rate is rising annually, and treatment costs are substantially higher when infections are present, with variations depending on country and healthcare system [3]. Research has shown that foot ulcers account for 85% of diabetic lower limb amputations [4].

In Vietnam, diabetic foot disease is a major source of morbidity among patients with T2DM. It is estimated that around 5% of the diabetic population will experience a foot ulcer, and 1.5% have an active ulcer at any given time. The global lifetime prevalence of diabetic foot ulcers is 6.3%, and each year, approximately 1–4.1% of diabetic patients develop new ulcers [5]. Over the course of a diabetic patient's lifetime, the risk of developing foot complications is 25% [6]. The consequences of foot ulcer complications are significant: the rate of disability is high, patients lose the ability to work, treatment is complicated and expensive, and the recurrence rate increases with the duration of the illness. These factors strongly affect the lives of patients, their families, society, and the healthcare system [7, 8]. Globally, over one million lower limbs are amputated each year due to diabetes-related complications, accounting for about 70% of all non-traumatic amputations. The risk of amputation is 15 times higher in people with diabetes. Approximately half of all diabetic-related amputations are major (below or above the knee), while the rest are minor (toes or parts of the foot).

In Vietnam, farmers and patients living in rural areas often experience more severe foot complications and face higher treatment costs due to delayed healthcare access and lower health literacy. Their per capita income is significantly lower than that of urban residents, which increases the burden of care [9].

Despite the significant impact of diabetic foot ulcers, statistics on their clinical characteristics and related expenditures in Vietnam are limited. This study aims to fill this gap by identifying the clinical features of diabetic foot ulcers in patients with T2DM in northern Vietnamese hospitals, quantifying associated healthcare costs, and analyzing socioeconomic factors affecting disease severity and treatment expenses—particularly in rural populations. Additionally, the study offers policy recommendations aimed at improving access to care and reducing the economic burden for low-income patients.

Methods

Study design, time, and setting

This study was designed as a cross-sectional, descriptive Cost of Illness (COI) analysis, focusing specifically on direct hospital charges from the patient's perspective. A total of 283 patients hospitalized with diabetic foot ulcers were prospectively enrolled between June 2016 and September 2019 at five general hospitals in Hanoi, Vietnam. These included two provincial-level hospitals (Thanh Nhan and Saint Paul General Hospitals) and three central-level hospitals (the National Hospital of Endocrinology, Hanoi Medical University Hospital, and the National Institute of Diabetes and Metabolic Disorders). All participating hospitals were equipped with diabetes specialists and microbiology laboratories and served both urban and rural populations in Hanoi and surrounding provinces.

Inclusion and exclusion criteria

Eligible participants were aged ≥ 18 years, had type 2 diabetes, and presented with new-onset foot ulcer(s) or gangrene below the ankle. Recurrent ulcers were included only if previously healed (complete epithelialization without discharge). For patients with multiple ulcers, the largest in diameter or depth was recorded [10].

Patients with foot ulcers due to trauma, venous stasis, or connective tissue diseases were excluded.

Data collection

Trained researchers conducted structured interviews and reviewed medical records. Collected data included demographic characteristics (age, sex, residence, occupation, income), clinical history, years with diabetes, diabetes treatment, and ulcer duration. Laboratory tests collected fasting plasma glucose (FPG), random plasma glucose (RPG), and HbA1c levels. Peripheral neuropathy was assessed through vibration perception, muscle strength, and tendon reflex testing. Pressure, vibration, pain, and proprioception were evaluated bilaterally using standardized tools. Structural deformities included claw toes, hammer toes, hallux valgus, Charcot foot, and others [11, 12].

Diabetic foot ulcer classification

Foot ulcers were graded using the Wagner classification [11]:

Grade 0: Intact skin with structural deformities.
Grade 1: Superficial ulcers.
Grade 2: Deep ulcers without bone involvement.
Grade 3: Deep ulcers with abscess/osteomyelitis.
Grade 4: Partial foot gangrene.
Grade 5: Whole foot gangrene.

Table 1 Demographic and clinical characteristics of diabetic foot ulcer patients

Characteristics	*n* = 283	%
Demographic characteristics		
Mean age (years)	62.54 ± 12.75	-
Age groups (years)		
< 30	4	1.4
30–39	9	3.2
40–49	25	8.8
50–59	78	27.6
60–69	87	30.7
≥ 70	80	28.3
Gender		
Men	123	43.5
Women	160	56.5
Residence		
Urban	136	48.1
Rural	147	51.9
Occupation		
Government employee	12	4.2
Worker	12	4.2
Trader	4	1.4
Self-employed	23	8.1
Farmer	79	27.9
Retired	105	37.1
Others	48	17.0
Duration of diabetes		
Mean duration (years)	9.29 ± 6.24 (1–30)	-
< 5 years	70	24.7
5–10 years	124	43.9
> 10 years	89	31.4
Biochemical characteristics		
Fasting plasma glucose (mmol/L)	8.87 ± 2.79	-
Random plasma glucose (mmol/L)	14.41 ± 4.36	-
HbA1c (%)	8.94 ± 2.23	-

Abbreviations: RPG – Random plasma glucose; HbA1c – Glycated hemoglobin

Antibiogram

In cases of clinical infection, wound cultures were obtained and antibiograms were performed. Wounds were debrided and swabbed deeply, with samples sent for Gram stain, culture, and antimicrobial susceptibility testing according to standard protocols [10, 13, 14].

Length of hospital stay (LOS)

LOS was defined as the number of days from admission to discharge and included only days related to diabetic foot ulcer care.

Cost measurement

Only direct medical costs (hospital charges) were included, measured from the patient's perspective. These covered expenses for diagnostics (e.g., X-ray, blood/urine tests), hospitalization (bed-days), wound care (dressings and debridement), and medications (antibiotics, insulin, and oral hypoglycemic agents). Indirect costs (e.g., lost income, caregiver burden) and societal opportunity costs were not included. Unit costs and quantities were averaged across the patient sample (n = 283), and all values were standardized to 2019 VND using Vietnam's health-related Consumer price index (Statista, 2024) [15]. A breakdown of cost components is shown in Table 1.

Outcomes

Primary: Total direct medical cost (in 2019 VND). Rural vs. urban cost differences were analyzed using bivariate tests and a generalized linear model (GLM), controlling for age, sex, occupation, income, HbA1c, Wagner grade, ulcer duration, and bacterial infection status.

Secondary: Length of hospital stay (LOS), similarly compared between rural and urban patients. Stratified tables and adjusted models were used to isolate rural effects.

Statistical analysis

Quantitative variables were tested for normality and compared using ANOVA or t-tests (for normally distributed data), and Mann–Whitney U or Kruskal–Wallis tests (for non-normal data). Categorical variables were analyzed using chi-square or Fisher's exact test. Results were expressed as means ± standard deviations (SDs), medians, or odds ratios (ORs) with 95% confidence intervals (CIs).

Multivariable analysis employed GLM with gamma distribution and log link to model log-transformed treatment costs. Predictors included age, sex, rural residence, ulcer grade, HbA1c, infection type (e.g., Acinetobacter baumannii, Pseudomonas aeruginosa), and LOS.

Table 2 Characteristics of diabetic foot ulcers ($n = 283$)

Category	n	%
Bacterial culture results		
Positive culture	21	80.8
Negative culture	5	19.2
Bacterial species identified (among positives)		
Staphylococcus aureus	6	28.6
E. coli	5	23.8
Enterococci spp.	5	23.8
Proteus mirabilis	4	19.0
Pseudomonas aeruginosa	2	9.5
Acinetobacter baumannii	2	9.5
Enterobacter cloacae	1	4.8
Burkholderia cepacia	1	4.8
Klebsiella pneumoniae	1	4.8
Stenotrophomonas maltophilia	1	4.8
Streptococcus agalactiae	1	4.8
Ulcer classification (Wagner)		
Grade 0	36	12.7
Grade I	34	12.0
Grade II	86	30.4
Grade III	82	29.0
Grade IV	38	13.4
Grade V	7	2.5
Ulcer location		
Toes (general)	86	30.4
Intertoe spaces	21	7.4
Big toe (hallux)	18	6.4
Little toe	21	7.4
Instep	27	9.5
Insole	29	10.2
Heel	27	9.5
Lateral/side of foot	16	5.7
Amputation stump	16	5.7
Cause of ulcer		
Foreign objects	63	22.3
Blisters	50	17.7
Tight shoes	18	6.3
Burns	18	6.4
Corns	17	6.0
Improper toenail trimming	12	4.2
Unknown	81	28.6
Other	24	8.5
Associated risk factors		
Sensory disturbances	250	88.3
Loss of tendon reflexes	251	88.6
History of previous foot ulcer	132	46.6
History of amputation	63	22.3
Foot deformity	50	17.6

E. coli: Escherichia coli, B. cepacia: Burkholderia cepacia

Model formula:

$$log\,(Cost) = \beta_0 + \beta_1\,(Rural) + \beta_2\,(Wagner\,Grade) + \beta_3\,(Infection\,Type) + \beta_4\,(LOS) + \beta_s\,(HbA1c) + \varepsilon$$

LOS was modeled similarly, excluding cost. Missing data (< 5%) were handled via complete-case analysis, and multiple imputation was considered in sensitivity checks.

Statistical significance was set at $p < 0.05$. Analyses were conducted using SPSS version 16.0.

Ethics

The study adhered to the Declaration of Helsinki. Written informed consent was obtained from all participants. Ethical approval was granted by the National Institute of Diabetes and Metabolic Disorders, Hanoi Medical University (Approval No. 126/QĐ-ĐTĐ; Date: 13 October 2015).

Results

A total of 283 patients were included in the study, with a higher proportion of females (56.5%) (Table 1). The mean age was 62.54 ± 12.75 years, and approximately 87% of participants were aged over 50 years. The distribution of patients by residence was relatively balanced, with 51.9% from rural areas and 48.1% from urban areas. The majority of patients were farmers (27.9%) or retired (37.1%), reflecting a predominantly low-income population. The mean duration of diabetes was 9.29 ± 6.24 years, ranging from 1 to 30 years. Nearly 44% of patients developed foot ulcers after 5–10 years of living with diabetes. Biochemical analysis indicated poor glycemic control among participants: the mean fasting plasma glucose was 8.87 ± 2.79 mmol/L, random plasma glucose was 14.41 ± 4.36 mmol/L, and HbA1c was 8.94 ± 2.23%. Notably, the proportion of farmers and retirees (65%) was significantly higher than that of other occupational groups, underscoring the socioeconomic vulnerability of patients with diabetic foot ulcers.

Among the 283 patients with diabetic foot ulcers, 80.8% (21/26) of wound cultures tested positive for bacterial infection, with Gram-negative bacteria accounting for 76.2% of isolates (Table 2). The most frequently identified species were *Staphylococcus aureus* (28.6%), *E. coli* and *Enterococci* (23.8% each), followed by *Proteus mirabilis* (19%) and *Pseudomonas aeruginosa* (9.5%). Other detected organisms included *Acinetobacter baumannii*, *Klebsiella pneumoniae*, and *Burkholderia cepacia*. Regarding ulcer severity, Grade II (30.4%) and Grade III (29.0%) ulcers were most common based on Wagner classification. Notably, 44.9% of patients had ulcers classified as Grade III or higher, and 15.9% had severe ulcers (Grade IV–V), indicating delayed presentation and

Table 3 Relationships between blood glucose parameters, hospital stay duration, and patient characteristics

Variable	Group	*n*	Mean ± SD	*p*-value
Fasting Glucose (mmol/L)				
Diabetes duration < 5 years		70	8.29 ± 2.04	p1–2 = 0.218
5–10 years		124	8.90 ± 2.73	p1–3 = 0.104
> 10 years		89	9.13 ± 2.98	p2–3 = 0.916
HbA1c (%)				
Diabetes duration < 5 years		63	8.85 ± 2.24	p1–2 = 0.807
5–10 years		111	8.77 ± 2.21	p1–3 = 0.417
> 10 years		81	9.14 ± 2.15	p2–3 = 0.232
HbA1c by Residence				
Urban		123	8.56 ± 1.89	0.011
Rural		132	9.23 ± 2.33	
HbA1c by Occupation				
Government employee		11	8.59 ± 2.16	0.624
Worker		12	9.02 ± 1.58	
Trader		4	8.13 ± 1.70	
Self-employed		21	9.30 ± 2.75	
Farmer		66	9.12 ± 2.36	
Retired		99	8.62 ± 1.91	
Other		42	9.16 ± 2.20	
Hospital Stay (days)				
Age < 30		4	11.00 ± 6.78	0.052
30–39		9	13.33 ± 6.59	
40–49		25	13.64 ± 11.02	
50–59		78	12.77 ± 11.50	
60–69		87	11.99 ± 11.89	
≥ 70		80	8.98 ± 7.30	
Hospital Stay by Bacteria				
Staphylococcus aureus		6	23.83 ± 19.17	0.506
E. coli		5	6.40 ± 4.83	
Enterococci		5	21.80 ± 11.86	
Proteus mirabilis		4	21.75 ± 15.15	
A. baumannii		2	38.00 ± 22.63	
P. aeruginosa		2	36.50 ± 31.82	
Others (combined)		5	Variable	

SD: standard deviation

advanced tissue damage. The most frequent ulcer sites were the toes (30.4%) and instep (24%). Recurrent ulcers at previous amputation stumps accounted for 5.7%. The leading causes of ulcers were trauma from foreign objects (22.3%) and blistering (17.7%), while 28.6% had no clearly identified cause. Neuropathy was highly prevalent, with 88.3% experiencing sensory disturbance and 88.6% showing reduced tendon reflexes. Prior foot ulcers (46.6%) and previous amputations (22.3%) were common, highlighting the chronic and recurrent nature of diabetic foot complications.

As shown in Table 3, both fasting glucose and HbA1c levels increased with longer diabetes duration; however, the differences were not statistically significant ($p > 0.05$). Patients with diabetes for over 10 years had the highest average glucose (9.13 ± 2.98 mmol/L) and HbA1c levels (9.14 ± 2.15%). Patients from rural areas had significantly poorer glycemic control than those from urban areas, with higher mean HbA1c values (9.23 ± 2.33% vs. 8.56 ± 1.89%, $p = 0.011$). Although traders and government employees had relatively lower HbA1c levels, no significant differences were found across occupational groups ($p = 0.624$). Hospital stay duration varied by age and infecting bacteria. The longest average stays were observed in patients aged 40–49 (13.64 ± 11.02 days) and 30–39 (13.33 ± 6.59 days). The difference in stay duration among age groups approached significance ($p = 0.052$). Among infected patients, those with *Acinetobacter baumannii* and *Pseudomonas aeruginosa* experienced the longest hospitalizations (38.0 ± 22.63 and 36.5 ± 31.82 days, respectively), though the differences across bacterial species were not statistically significant ($p = 0.506$).

The total direct medical cost per patient per hospitalization was estimated at approximately 10.5 million VND in 2019 values (Table 4). The largest contributors to treatment costs were antibiotic therapy (mean: 8.5 million VND) and bed-days (4.8 million VND), followed by wound care procedures (2.4 million VND). Diagnostic services (including laboratory tests and imaging) accounted for around 2.15 million VND, while medications such as insulin and oral hypoglycemic agents contributed approximately 2.9 million VND per patient. These findings underscore the substantial cost burden associated with infection management and prolonged hospitalization in diabetic foot ulcer treatment.

Table 5 presents a comparison of clinical features and treatment outcomes between rural and urban patients. Glycemic control was modestly better in rural patients, with a statistically significant lower HbA1c (9.23 ± 2.33 vs. 9.56 ± 1.89, $p = 0.0407$). However, no significant differences were observed between the two groups in terms of ulcer severity, length of hospital stay, or treatment cost.

Table 4 Breakdown of direct medical costs by component in diabetic foot ulcer patients (2019 VND, $n = 283$)

Cost Component	Unit Cost (2019 VND)	Quantity (Mean per Patient)	Total Contribution (Mean, 2019 VND)
Diagnostics (X-ray, labs)	500,000–1,000,000	3–5 tests	2,150,000
Bed-days (LOS-related)	1,200,000/day	12.5 days	4,800,000
Dressings/debridement	300,000/session	8 sessions	2,400,000
Antibiotics	2,000,000–5,000,000/course	1–2 courses	8,500,000
Insulin/OHA/Other medications	500,000–1,500,000	Variable	2,900,000
Total Estimated Direct Cost	-	-	**10,500,000**

Table 5 Comparison of clinical characteristics and treatment outcomes between rural and urban patients

Variable	Urban ($n = 147$)	Rural ($n = 136$)	*p*-value*
HbA1c (%)	9.56 ± 1.89	9.23 ± 2.33	0.0407
Ulcer severity (Wagner grade)	2.31 ± 1.31	2.19 ± 1.23	> 0.05
Length of hospital stay (days)	10.83 ± 10.07	12.16 ± 10.77	> 0.05
Treatment cost (thousand VND)	9,444 ± 7,239	10,154 ± 6,967	> 0.05

*Mann-Whitney U test

These results suggest that, despite potential disparities in healthcare access, rural and urban patients experienced comparable clinical severity and resource utilization during hospitalization for diabetic foot ulcers.

Table 6 presents the results of a multivariable regression analysis using a generalized linear model (GLM) with gamma distribution. After adjusting for relevant covariates, two factors were independently associated with increased treatment costs: Positive bacterial culture ($p = 0.002$); Higher Wagner ulcer grade ($p = 0.038$). In contrast, other variables such as age, occupation, region, HbA1c, and duration of diabetes were not statistically significant predictors of cost. These findings underscore the importance of infection and ulcer severity as key drivers of economic burden among hospitalized patients with diabetic foot ulcers.

Discussion

Characteristics of diabetic foot ulcers

Most patients with diabetic foot ulcers were older than 50 years (86.6%), consistent with prior research indicating increased vulnerability in older adults. Compared to a 2008 study where the 51–70 age group accounted for 56.3% of patients [13], our findings suggest a trend toward an aging population with prolonged diabetes duration and inadequate glucose control.

We observed a higher proportion of female participants (56.5%), contrasting with some international studies that report male predominance [16, 17]. In Vietnam, women—particularly in rural areas—may have less access to preventive care and seek treatment only after complications arise. This behavioral and systemic difference may explain the gender shift in our population.

The prevalence of foot ulcers was similar in rural and urban areas; however, patients in rural areas face more limited access to specialist care, predisposing them to higher risks of amputation and mortality, similar to findings by Sutherland et al. [18]. In our study, the proportion of farmers and retirees was significantly higher, which may be linked to barefoot walking habits and delayed medical attention, as noted in previous literature.

Table 6 Multivariable regression analysis of factors associated with total treatment cost among patients with diabetic foot ulcers ($n = 283$)

Variable	Coefficient	Std. error	z	*p*-value	95% Confidence interval
Age (years)	-0.006	0.004	-1.44	0.151	[-0.014, 0.002]
Occupation					
Government officer (Ref.)	-	-	-	-	-
Worker	0.122	0.405	0.30	0.763	[-0.671, 0.916]
Small trader	-0.156	0.440	-0.36	0.722	[-1.018, 0.705]
Self-employed	0.118	0.299	0.40	0.692	[-0.467, 0.703]
Farmer	0.187	0.254	0.74	0.462	[-0.311, 0.686]
Retired	-0.126	0.261	-0.48	0.629	[-0.637, 0.385]
Other	-0.284	0.283	-1.00	0.315	[-0.839, 0.270]
Region					
Urban (Ref.)	-	-	-	-	-
Rural	-0.070	0.094	-0.75	0.453	[-0.254, 0.113]
HbA1c (%)	0.004	0.021	0.17	0.868	[-0.038, 0.046]
Ulcer severity (Wagner grade)	0.091	0.044	2.08	0.038*	[0.005, 0.176]
Culture result					
Negative (Ref.)	-	-	-	-	-
Positive	0.381	0.125	3.04	0.002*	[0.136, 0.626]
Duration of diabetes (years)	-0.002	0.007	-0.27	0.788	[-0.016, 0.012]
Constant (intercept)	16.259	0.426	38.18	< 0.001	[15.424, 17.093]

* Statistically significant at $p < 0.05$

Model: Generalized Linear Model (GLM), Gamma distribution, log link

Model deviance = 91.95; scale parameter = 0.47

A GLM with gamma distribution and log link was applied. Coefficients represent the log-transformed effect on cost. A positive bacterial culture and higher Wagner grade were significantly associated with increased hospitalization costs

Foot ulcers typically developed 5 to 10 years after diabetes diagnosis, with a mean duration of diabetes being 9.29 years. This matches trends reported by Yazdanpanah et al. [19], suggesting a mid-to-late disease complication timeline. Persistent hyperglycemia and poor glycemic control were prominent in our cohort, confirming associations with diabetic complications, as seen in Bansal et al. [13]. High glucose levels impair microvascular circulation and increase the risk of skin breakdown and infection, particularly in distal extremities.

Notably, rural patients and those in self-employed or farming occupations had worse HbA1c levels, potentially due to financial constraints or limited healthcare access. Others studies confirm the role of socioeconomic status in diabetes outcomes [20, 21].

Patients with Acinetobacter baumannii and Pseudomonas aeruginosa infections had the longest hospital stays, mirroring trends in Kim et al. [22] and other research linking bacterial resistance to prolonged hospitalization [23, 24]. These pathogens are frequently isolated in severe infections requiring extended care.

In our study, most ulcers were classified as Wagner Grades 2 and 3, consistent with delayed diagnosis and treatment. Ulcer location was most often at the toes and instep, reflecting neuropathy-induced pressure insensitivity in these areas [25, 26]. Patients with prior amputations or structural deformities were more likely to have recurrence, as highlighted by Armstrong et al. [7]. Proper footwear, hygiene, and routine foot care remain key preventive strategies.

Peripheral neuropathy was the most common underlying condition (88.3%), consistent with other reports [27]. Even after vascular interventions, ulcer recurrence is common due to persistent neuropathy, reinforcing the need for lifelong follow-up and patient education [11].

Expenditures for diabetic foot ulcers

The average direct medical cost per patient was approximately VND 9.2 million (US$458.8), with major contributors being antibiotics, hospital bed costs, and wound care. Costs increased with ulcer severity (Wagner Grades 3–5), prolonged hospitalization, and infection, aligning with findings from Driver et al. [28, 29].

The highest costs were observed in patients aged 40–49 years, although the differences were not statistically significant between age groups. Male patients and farmers had higher expenditures, likely due to delayed presentation and advanced disease, paralleling earlier findings [30, 31].

Rural patients had significantly higher average treatment costs compared to urban patients. As found in prior studies [18], disparities in healthcare access in rural areas often result in delayed diagnosis, greater complications, and higher treatment burden.

Our findings reinforce that higher Wagner grades and positive bacterial cultures independently predict increased treatment costs, highlighting the need for early intervention and infection control measures. This is consistent with previous cost-effectiveness analyses in diabetic foot ulcer care [32, 33].

The findings emphasize the need for early detection and referral of diabetic foot ulcers, particularly in rural settings. Strengthening primary care screening, expediting referrals, and subsidizing treatment costs for rural patients could help reduce the burden of severe ulcers. A multidisciplinary care model may be effective in improving outcomes and optimizing costs.

Limitations

This study was limited to hospitalized patients in northern Vietnam, which may affect generalizability to other regions. Long-term outcomes, such as ulcer recurrence and amputation rates, were not assessed. Additionally, comparative analysis with non-ulcer diabetic patients was not performed. These limitations suggest avenues for future research on prevention strategies, long-term costs, and broader population impact.

Conclusion

Patients with diabetic foot ulcers living in rural areas demonstrated poorer glycemic control, longer hospital stays, and incurred higher treatment costs than those living in urban areas. Higher-grade lesions, as classified by the Wagner system, were significantly associated with increased treatment expenditures, largely due to extended hospital stays, more intensive use of medications, and additional diagnostic and procedural interventions. Infections with multidrug-resistant organisms, particularly *Acinetobacter baumannii* and *Pseudomonas aeruginosa*, were linked to the longest hospitalizations and the highest costs, highlighting the burden of antimicrobial resistance in diabetic foot care.

Abbreviations

BMI	Body Mass Index
IFG	Impaired Fasting Glucose
RPG	Random Plasma Glucose
LEA	Lower Extremity Amputation
T2DM	Type 2 Diabetes Mellitus

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Author contributions
DDT, TVB, and NTX conceptualized the study design. TNT and NTL independently collected and extracted the data. NDL, TNT, and NTL verified and supplemented the data under the guidance of senior authors. DDT prepared the first draft of the manuscript. All authors (DDT, TVB, NDL, TNT, NTL, NTX) contributed to data interpretation, manuscript review, and approval of the final version.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
The study was approved by the National Institute of Diabetes and Metabolic Disorders, Hanoi Medical University (Approval No. 126/QĐ-ĐTĐ). Informed written consent was obtained from all participants prior to enrollment. Participants were assured of the confidentiality of their medical information.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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